

**B.M.S COLLEGE OF ENGINEERING BENGALURU**  
**Autonomous Institute, Affiliated to VTU**



**PLC AAT Report on**

**WEATHER APP**

*Submitted in partial fulfillment of the requirements of AAT*

Bachelor of Engineering in  
Computer Science and Engineering

*Submitted by:*

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2025-2026

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



### **DECLARATION**

We, Vidhan Tewatia, Tushar Garg, Vishesh Aggarwal, and Varun, students of the 2nd Semester, B.E., Department of Computer Science and Engineering, BMS College of Engineering, Bangalore, hereby declare that this AAT Project entitled “**WEATHER APP**” has been carried out in the Department of CSE, BMS College of Engineering, Bangalore during the academic semester February 2026 – July 2026.

#### **Student Name**

#### **Student Signature**

1. *Vidhan Tewatia*

2. *Tushar Garg*

3. *Vishesh Aggarwal*

4. *Varun*

**BMS COLLEGE OF ENGINEERING**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**



***CERTIFICATE***

This is to certify that AAT project titled 'WEATHER APP' has been carried out by :  
VIDHAN TEWATIA (1BM25CS313)  
TUSHAR GARG (1BM25CS305)  
VARUN (1WM25CS294)  
VISHESH AGGARWAL (1BM25CS314)  
during academic year 2025-2026

Faculty Signature

## Table of Contents

| Sl. No. | Title                 | Page no. |
|---------|-----------------------|----------|
| 1       | Introduction          | 1        |
| 2       | Algorithm             | 2        |
| 3       | Flowchart             | 3        |
| 4       | Source code           | 4        |
| 5       | Results (screenshots) | 5        |
| 6       | Refrences             | 6        |

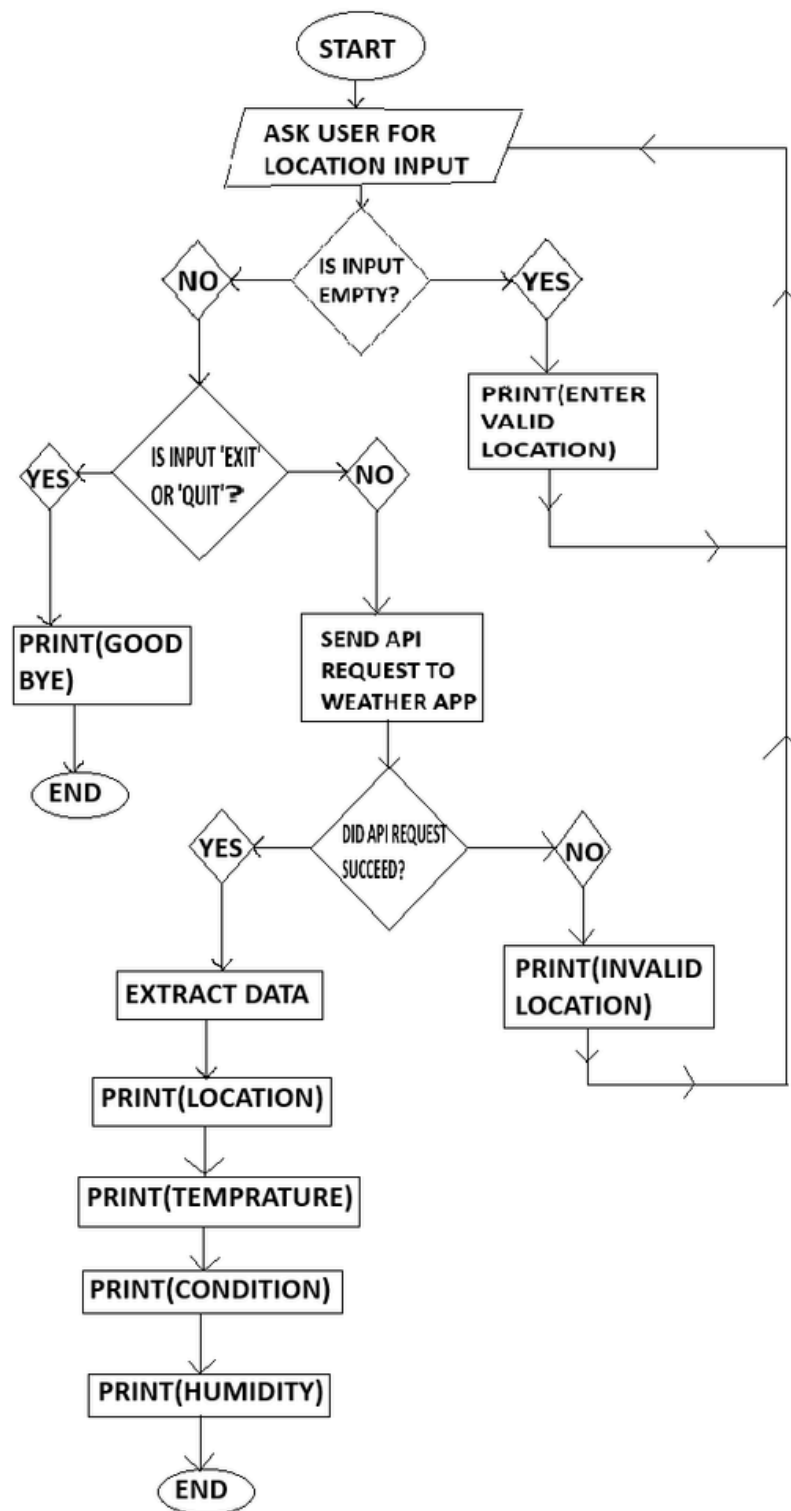
## INTRODUCTION

Weather plays a crucial role in our daily lives, influencing decisions ranging from travel plans to agricultural activities. With the increasing reliance on technology, mobile and web applications have become essential tools for accessing real-time weather information. This project focuses on the development of a Python-based weather application designed to provide users with accurate and timely weather updates. By integrating APIs and leveraging Python's versatility, the app delivers key details such as temperature, humidity, wind speed, and forecasts in a user-friendly interface. The primary objective of this project was not only to enhance technical skills in Python programming but also to demonstrate the practical application of software development in solving everyday problems. Through this project, I explored concepts such as API handling, data visualization, and error management, ultimately creating a functional and accessible weather app.

# ALGORITHM

1. Start
2. Initialize Constants:
  - a.1. Set **`API\_KEY`**.
  - b.2. Set **`BASE\_URL`** to the Weather API endpoint.
3. Begin Infinite Loop:
  - a.1. Repeat until the program is manually stopped.
4. Take User Input:
  - a.1. Prompt the user: *"Enter a location."*
  - b.2. Store the input in **`location`**.
5. Validate Input:
  - a.1. Check if **`location`** is empty.
  - b.1.1. If empty, display *"Please enter a valid location."*
  - c.1.2. Go back to Step 4.
6. Prepare API Request:
  - a.1. Create a **`params`** dictionary:
  - b.1.1. **`key`** = **`API\_KEY`**.
  - c.1.2. **`q`** = **`location`**.
  - d.2. Send request using **`requests.get(BASE\_URL, params=params)`**.
  - e.3. Store the response.
7. Check Response Status:
  - a.1. If **`status\_code == 200`**:
    - b.1.1. Parse JSON data.
    - c.1.2. Extract:
      - d.1.2.1. Location name and country.
      - e.1.2.2. Temperature (°C).
      - f.1.2.3. Weather condition.
      - g.1.2.4. Humidity.
    - h.1.3. Display formatted weather details.
  - i.2. Else:
    - j.2.1. Display *"Please enter a valid location."*
8. Repeat Loop:
  - a.1. Return to Step 4.
9. End (Only when the program is terminated manually).

# FLOWCHART



## SOURCE CODE

```
import requests

API_KEY = <API_KEY>
BASE_URL = "https://api.weatherapi.com/v1/current.json"

while True:
    location = input("Enter a location: ").strip()

    if not location:
        print("Please enter a valid location.")
        continue

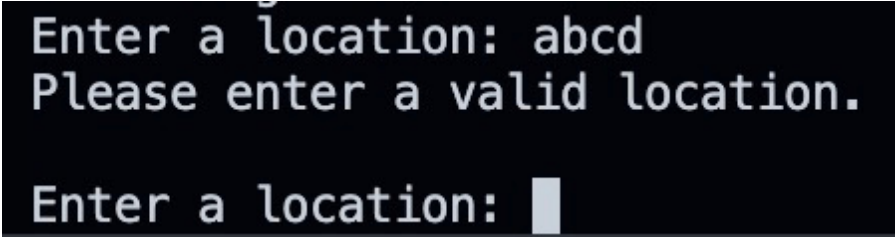
    params = {"key": API_KEY, "q": location}
    response = requests.get(BASE_URL, params=params)

    if response.status_code == 200:
        data = response.json()
        loc = data["location"]
        current = data["current"]

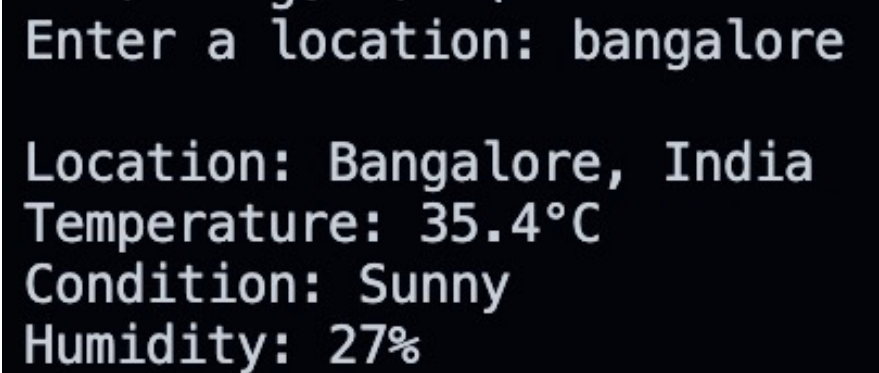
        print(f"\nLocation: {loc['name']}, {loc['country']}")
        print(f"Temperature: {current['temp_c']}°C")
        print(f"Condition: {current['condition']['text']}")
        print(f"Humidity: {current['humidity']}%\n")
    else:
        print("Please enter a valid location.\n")
```



## RESULTS (SCREESHOTS)

1) A terminal window with a black background and white text. The first line says "Enter a location: abcd". The second line says "Please enter a valid location.". The third line says "Enter a location:" followed by a white cursor bar.

```
Enter a location: abcd
Please enter a valid location.
Enter a location: █
```

2) A terminal window with a black background and white text. The first line says "Enter a location: bangalore". The second line says "Location: Bangalore, India". The third line says "Temperature: 35.4°C". The fourth line says "Condition: Sunny". The fifth line says "Humidity: 27%".

```
Enter a location: bangalore
Location: Bangalore, India
Temperature: 35.4°C
Condition: Sunny
Humidity: 27%
```

# REFERENCES

[1] WeatherAPI, "Weather API Documentation," [Online]. Available: <https://www.weatherapi.com/docs/>

[2] Python Software Foundation, "Python Documentation," [Online]. Available: <https://docs.python.org/3/>

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[5] REST API Tutorial, "RESTful Web Services," [Online]. Available: <https://restfulapi.net/>